

CAMPULUNG MUSCEL, RomaniaWMO: **153240**Lat: **45.2747N**Lon: **25.0367E**Elev: **682**StdP: **93.39**Time Zone: **2.00 (EUE)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-9.7	-15.2	1.1	-8.4	-13.4	1.3	-6.9	10.0	-2.9	8.0	-3.2	1.2	50	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	28.4	19.5	27.1	18.8	25.8	18.3	20.6	26.9	19.7	25.5	19.0	24.6	1.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	14.5	23.5	17.6	13.7	22.7	16.9	13.1	21.9	62.6	27.1	59.4	25.4	56.7	24.6	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	5.3	4.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-1.5	1.3	4.0	8.5	13.6	17.6	18.5	18.9	14.2	9.4	4.2	0.0
	DBStd	8.10	4.35	4.30	4.41	4.32	3.07	2.92	2.66	2.77	3.56	4.31	4.35	4.23
	HDD10.0	1436	358	244	191	78	8	0	0	0	7	63	178	310
	HDD18.3	3483	616	477	444	296	149	46	31	27	129	277	423	568
	CDD10.0	1108	0	0	5	32	118	229	264	276	133	45	6	0
	CDD18.3	113	0	0	0	0	1	26	37	45	5	0	0	0
	CDH23.3	1031	0	0	0	3	25	231	319	392	58	3	0	0
	CDH26.7	169	0	0	0	0	1	38	54	73	3	0	0	0
Wind	WSAvg	1.2	1.1	1.2	1.5	1.5	1.3	1.3	1.3	1.1	1.2	1.0	1.0	1.0
Precipitation	PrecAvg	853	41	39	49	72	102	115	113	89	70	63	52	51
	PrecMax	1165	76	73	84	138	179	202	198	185	162	143	121	103
	PrecMin	533	7	7	9	31	46	45	48	27	13	3	5	8
	PrecStd	164	21	19	22	27	40	44	44	42	38	34	28	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	15.3	19.7	23.3	25.8	30.3	30.2	30.4	27.1	23.6	18.0	13.6
		MCWB	5.9	9.4	10.7	13.7	17.0	20.9	21.3	20.5	16.8	14.8	10.9	7.1
	2%	DB	9.2	12.4	17.1	21.0	23.7	27.7	28.1	28.4	24.9	21.4	15.0	9.6
		MCWB	4.4	7.1	9.6	12.9	16.0	19.6	20.2	18.9	16.0	13.3	10.2	5.6
	5%	DB	6.3	9.8	14.1	18.5	22.0	26.0	26.6	27.1	22.7	19.1	12.7	7.6
		MCWB	3.2	5.4	8.0	11.1	15.2	18.9	19.2	18.5	15.5	12.8	9.2	5.0
	10%	DB	4.1	7.7	11.3	16.0	20.3	24.4	25.0	25.5	20.9	16.3	10.1	5.4
		MCWB	2.0	4.1	6.4	10.1	14.2	18.1	18.4	18.2	14.7	11.5	7.7	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	9.7	11.9	14.6	17.9	22.2	21.6	21.3	17.7	16.0	11.9	8.2
		MCDB	10.9	15.1	18.9	21.8	24.4	29.3	28.5	28.5	24.5	21.1	16.2	10.3
	2%	WB	4.9	7.6	10.1	13.3	16.7	20.3	20.5	20.2	16.9	14.3	10.9	6.7
		MCDB	8.0	11.7	16.8	19.9	22.7	26.4	26.6	26.3	23.3	19.2	14.3	9.3
	5%	WB	3.4	6.0	8.4	12.0	15.5	19.4	19.7	19.3	16.1	13.2	9.6	5.0
		MCDB	5.6	9.3	13.7	17.5	21.0	25.1	25.2	25.4	22.0	17.7	12.4	7.0
	10%	WB	2.1	4.7	6.9	10.8	14.6	18.6	18.8	18.6	15.3	12.2	8.2	3.7
		MCDB	3.6	7.3	11.0	15.4	19.5	23.9	23.9	24.5	20.3	15.5	9.9	5.4
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.9	9.6	10.2	10.2	11.0	11.2	11.5	10.8	10.1	8.0	7.2
		MCDBR	11.0	12.4	14.2	14.1	13.1	13.2	12.9	13.6	13.1	14.2	11.6	10.0
		MCWBR	6.9	7.9	8.2	7.3	6.7	6.7	6.5	6.1	6.0	7.4	6.7	6.4
	5% WB	MCDBR	10.0	11.7	13.6	13.0	12.4	12.6	11.7	12.4	12.4	12.5	10.3	9.0
		MCWBR	6.3	7.6	7.9	7.0	6.7	6.8	6.3	5.9	5.7	6.9	6.1	5.9
Clear-Sky Solar Irradiance	taub		0.304	0.336	0.379	0.440	0.419	0.420	0.434	0.451	0.404	0.366	0.339	0.308
	taud		2.323	2.241	2.197	2.073	2.204	2.239	2.206	2.150	2.254	2.354	2.374	2.364
	Ebn at Noon		796	831	843	818	851	850	832	799	808	791	748	752
	Edh at Noon		79	103	125	154	140	136	138	141	117	92	76	69
All-Sky Solar Radiation	RadAvg		1.40	2.23	3.33	4.37	5.33	5.79	5.88	5.29	3.87	2.65	1.53	1.15
	RadStd		0.16	0.30	0.35	0.47	0.50	0.50	0.48	0.42	0.36	0.34	0.16	0.14